

# Literature Review

## LLMs for Resume Tailoring

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April 29, 2026

### 1 Related Work Overview

Papers 1–5 focus on frameworks for resume generation or editing, while papers 6–8 examine frameworks for resume evaluation, often including suggestions for improvement. Paper 9 investigates the overall usefulness of resume generation in practice.

#### 1.1 Do LLM-Generated Resumes Make Me More Qualified? An Observational Study of LLMs For Resume Generation and Matching Tasks [9]

I decided to put this paper first because it provides an overview of this research field – generating and tailoring resumes to improve job-description matching. For the end user, the expected goal is to pass the initial resume screening. Employers increasingly use AI-based tools on their side, such as *Applicant Tracking Systems (ATS)*. On the job-seeker side, tailoring resumes has also begun to involve AI tools. The authors discuss and evaluate the main problems and negative effects of LLM use in the hiring process, such as *self-preference*, *algorithmic bias* against minorities or people with disabilities, and *resume whitening*. For example, the Claude model exhibited traces of resume whitening and removed some information from the skills section of the resume. Most importantly, the study finds that LLM-generated resumes tend to receive higher matching and ranking scores when evaluated by LLM-based systems, which reflects the type of screening performed by ATS tools.

##### **My Opinion:**

The paper provides a good introduction to the research topic. Furthermore, in the discussion section, it provides several useful guidelines for future research, such as using a multi-agent debate-based approach for resume rating and implementing an HCI-based approach to resume tailoring, where candidates are given the opportunity for selective tailoring.

#### 1.2 ResumeFlow: An LLM-facilitated Pipeline for Personalized Resume Generation and Refinement [5]

ResumeFlow is an open-source web tool (<https://github.com/Ztrimus/ResumeFlow>) that leverages LLMs to tailor a resume to a specific job posting. Rather than generat-

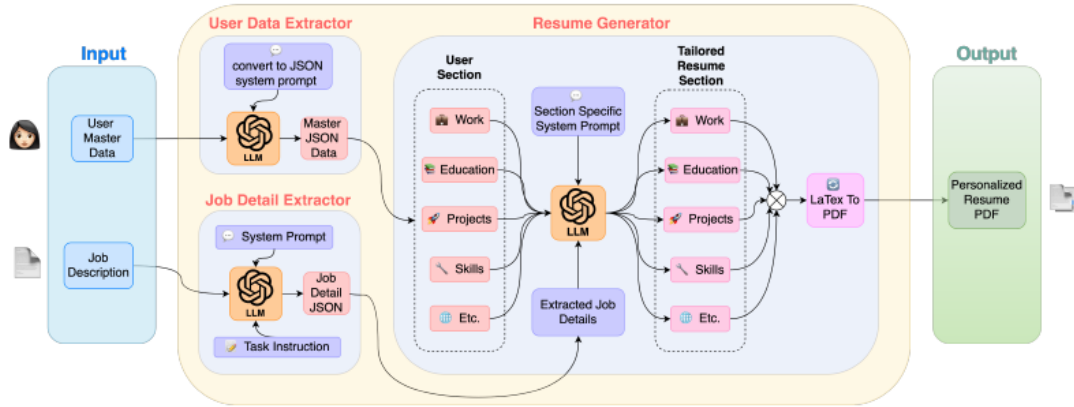


Figure 1: ResumeFlow Pipeline

ing a resume in a one shot, ResumeFlow breaks the process into stages: (1) user data extraction – converting an input PDF resume into JSON via a system prompt; (2) job details extraction – converting a copy-pasted job description into JSON via a system prompt, identifying key elements such as the title, keywords, purpose, responsibilities, qualifications, company name, and company information; and (3) resume generation – iterating over each section of the user data (the authors split it into five sections: Work, Education, Projects, Skills, and Etc.; see Fig. 1) and rewriting each section by retaining, emphasizing, or removing points to better match the target job. All three steps rely on LLM prompting. The authors show that this pipeline-based approach outperforms one-shot generation. They also introduce two evaluation metrics: *job alignment* and *content preservation*.

#### My opinion:

The paper is fairly basic, as it mainly presents a simple pipeline combined with prompt engineering. However, it is the first paper I have found (by publication date) that focuses on tailoring resumes. It provides code, useful baseline prompts, and practical insights about resume writing, such as best practices formulated by career counselors. I can also reuse the metrics introduced in this paper in my future research.

### 1.3 ApplicantAI: Transforming Resume Creation, Leveraging LLMs for Job Applications [1]

This is another, more recent work on leveraging LLMs for resume tailoring. To my understanding, it is conceptually the same as ResumeFlow, but with a slightly different pipeline and model choice. The system also takes two inputs – the user’s original resume and a job description – which are converted into JSON representations, after which the resume is re-generated using modified prompts. In particular, the authors employ Google’s recommended “XYZ” formula for crafting bullet points, encouraging the LLM to produce stronger, more results-oriented statements that are more likely to pass ATS screening. Their experiments show that using ApplicantAI is up to four times faster than manually rewriting a resume. However, they also observe that when the original resume is already well-written, the generated version can sometimes be slightly weaker, whereas less polished resumes tend to benefit more significantly.

### **My Opinion:**

This paper is less compelling than ResumeFlow. It is not open-source, and the authors do not introduce substantial novelty compared to ResumeFlow. Moreover, while primarily comparing their approach to ResumeFlow, they claim that ResumeFlow requires users to manually write prompts to generate a resume, which is inaccurate. The most valuable aspects of the paper are the directions for future work, such as integrating Optical Character Recognition (OCR) for resume parsing and enabling interactive user editing.

## **1.4 Enhancing job recommendations with LLM-based resume completion: A behavior-denoised alignment approach [3]**

The paper addresses the issue that job-seekers often do not update their resumes frequently, leading to gaps or outdated information that can reduce the effectiveness of job recommendation systems. The core solution involves using LLMs to automatically complete and enhance resume by generating content that is consistent with the user's previous work history and job preferences. The authors propose a new alignment method for fine-tuning LLM, called *Denoised Direct Preference Optimization (Denoised DPO)*, that allows to separate user preferences from noisy behavioral data. Their results showed that using their fine-tuned LLM improved job recommendations by increasing user engagement (7.52% more clicks per user) and conversion rates (13.29% more successful job applications).

### **My Opinion:**

Although the authors do not provide their dataset used for fine-tuning, the paper can be quite useful for my research. The approach of using LLMs to complete and update resumes based on user behavior (like job applications) aligns closely with the idea of dynamically adjusting a resume to better fit specific roles. Also their suggestion for future work to introduce knowledge graphs is what I plan to do.

## **1.5 Tailored Resume Generation Using RAG with LLM as per Job Specifications [4]**

The authors leverage *Retrieval-Augmented Generation (RAG)* with open-source LLMs to tailor resumes to specific job descriptions. In this setting, RAG serves as an external knowledge source that injects domain-relevant information, helping to constrain and guide the generated output. Additionally, the system allows users to iteratively refine the tailored resume through a conversational interface, where chatbot memory is used to preserve context across interactions. The overall architecture is illustrated in Fig. 2. The pipeline is broadly similar to ResumeFlow and ApplicantAI, but extends them by incorporating RAG. The input resume is first parsed and converted into JSON, extracting structured information for each section. This JSON follows a Pydantic-based schema, which is then provided to the LLM to generate the tailored resume. The paper concludes that LLM+RAG outperforms a standalone LLM in terms of cosine similarity for content preservation, achieving 89% compared to 72%, which is a key metric for evaluating tailored resumes.

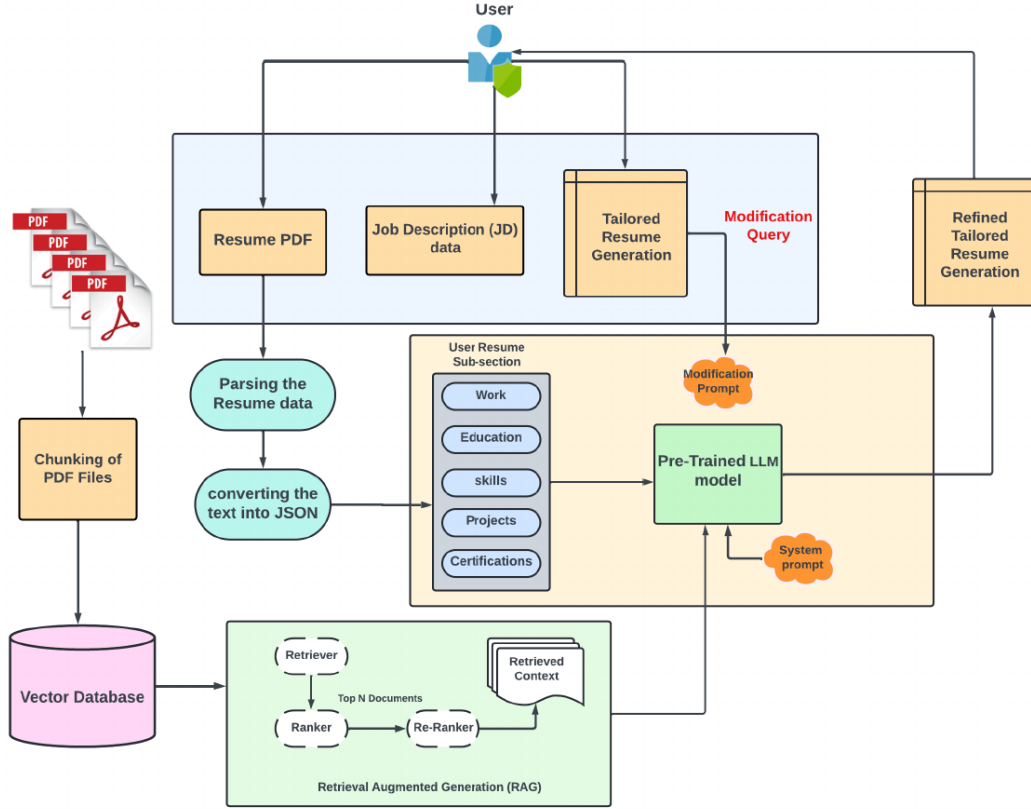


Figure 2: Architecture LLM + RAG

### My Opinion:

This paper serves as a strong candidate for a baseline in my work. It can be viewed as an extension of ResumeFlow and ApplicantAI, and provides a clear direction for future research that aligns with my ideas: (1) replacing RAG with RAFT (a hybrid of fine-tuning and retrieval), or (2) incorporating a knowledge graph. The authors also mention relevant re-ranking techniques and reference a dataset that could be useful for my experiments.

## 1.6 Ayna: a Novel LLM-Based Job Recruitment Plug-in [2]

I requested access to the paper but have not received it yet. Based on the abstract, the work appears to be quite insightful. The authors describe a Chrome extension built using FastAPI and LangChain, which may function as a complete agent-based system for automatically tailoring resumes and filling out job application forms.

## 2 Research Proposal

Traditional, general-purpose resumes are becoming less effective in today's highly competitive job market. This is particularly true for job seekers with limited work experience, as submitting a generic resume often leads to automatic rejection. This occurs because employers increasingly rely on automated screening systems, meaning that a resume may be filtered out before it even reaches a human reviewer. Many of these systems are AI-driven, such as *Applicant Tracking Systems (ATS)*. This highlights the importance

of tailoring a resume to a specific job posting in order to improve the chances of passing the screening stage. However, tailoring a resume manually typically takes around 15–20 minutes per application [1]. This process involves reading the job description, identifying relevant skills and responsibilities, and updating the resume accordingly. When applying to multiple positions, this repeated effort can lead to fatigue. Furthermore, applicants often struggle to identify trends, understand patterns, and consistently follow best practices in resume writing, which is itself a nuanced skill. These issues provide a strong motivation for developing automated solutions for resume tailoring.

Building a resume tailoring system is not just an engineering task, but also a research problem. Several challenges must be addressed through systematic research. These include: (1) identifying the most effective architecture — this requires developing solutions that outperform baselines (see the next paragraph) across multiple metrics, and ultimately selecting the best approach among candidates such as RAFT or knowledge graphs; (2) resume whitening / content preservation — where the LLM may remove key skills or projects that should be retained; (3) hallucination / fabrication — where the LLM introduces skills, projects, or other content that were not present in the original resume; (4) self-preference and algorithmic bias against minorities or people with disabilities — although these issues primarily arise on the employer’s side (during resume evaluation), they must also be considered from the applicant’s perspective; and (5) prompt design — careful prompt engineering is required to ensure that the LLM adheres to best practices in resume writing, such as the XYZ formula for bullet points and the use of active voice where appropriate. In the final pipeline, prompts may be used at multiple stages (not less than 3), each of which requires careful design.

Related work was reviewed in Section 1, where I summarized the core ideas of each paper and provided my own opinion. With the exception of one paper, all are conference publications and have been published within the last two years. This suggests that the area is both active and relatively underexplored. The three most relevant papers are [1], [4], and [5]. The first two, ApplicantAI and ResumeFlow, are relatively simple and primarily focus on prompt engineering. The third builds upon these approaches by using RAG, resulting in improved performance.

I propose to implement and compare several approaches, including a mixture of fine-tuning with RAG, as well as a knowledge graph-based solution. These methods will be evaluated against each other and against baselines to identify the most effective approach. The evaluation will consider metrics such as job alignment, content preservation, and standard NLP metrics including precision, recall, and F1 score (the final set of metrics is yet to be fully defined). The expected outcome of this research is to outperform baseline methods while also providing insights into the comparison between RAFT and knowledge graph-based architectures in the context of resume tailoring.

I plan to conduct this research and publish a paper, and I estimate that it can be completed within 1–1.5 months. Following this, I will build a multi-agent system enhanced with HCI components on top of the first paper (as I described in the previous report), which could become a second paper. Ultimately, the combined work will form the foundation of my master’s thesis, integrating both research and engineering contributions.

## References

- [1] D. Yan, M. Moh and T. -S. Moh  
*ApplicantAI: Transforming Resume Creation, Leveraging LLMs for Job Applications*, BigData 2025  
<https://ieeexplore.ieee.org/document/11401836>
- [2] Habib, Sukaina and Boaye Belle, Alvine  
*Ayna: a Novel LLM-Based Job Recruitment Plug-in*, LLM 2025  
[https://www.researchgate.net/publication/398124335\\_Ayna\\_a\\_Novel\\_LLM-Based\\_Job\\_Recruitment\\_Plug-in](https://www.researchgate.net/publication/398124335_Ayna_a_Novel_LLM-Based_Job_Recruitment_Plug-in)
- [3] Chen Zhu and Xiao Hu and Han Wu and Chuan Qin and Hengshu Zhu and Hui Xiong  
*Enhancing job recommendations with LLM-based resume completion: A behavior-denoised alignment approach*, (non-conference) 2025  
<https://www.sciencedirect.com/science/article/abs/pii/S030645732500202X>
- [4] Patra, Komal Prakashchandra and Diwakar, Manoj and Arya, Chandrakala  
*Tailored Resume Generation Using RAG with LLM as per Job Specifications*, PDGC 2024  
<https://ieeexplore.ieee.org/document/10984036>
- [5] Saurabh Bhausahab Zinjad and Amrita Bhattacharjee and Amey Bhilegaonkar and Huan Liu  
*ResumeFlow: An LLM-facilitated Pipeline for Personalized Resume Generation and Refinement*, ACM SIGIR 2024  
<https://dl.acm.org/doi/10.1145/3626772.3657680>
- [6] Awasthi, Dhruv and Rao, Pooja S. B. and Jayagopi, Dinesh Babu  
*ResumeGenAI: Supporting Job Seekers with LLM-Driven Resume Feedback*, ACM CUI 2025  
<https://dl.acm.org/doi/full/10.1145/3719160.3737622>
- [7] Vagale, Ghanashyam and Bhat, Shashidhar Y and Dharishini, P. Padma Priya and GK, Prajwal  
*ProspectCV: LLM-Based Advanced CV-JD Evaluation Platform*, SCES 2024  
<https://ieeexplore.ieee.org/document/10652548>
- [8] Sunico, Rafael Jace and Pachchigar, Shubh and Kumar, Vinay and Shah, Ishika and Wang, Jingyi and Song, Isabel  
*Resume Building Application based on LLM (Large Language Model)*, ICCIS 2023  
<https://ieeexplore.ieee.org/document/10425602>
- [9] S. Vaishampayan and C. Brown  
*Do LLM-Generated Resumes Make Me More Qualified? An Observational Study of LLMs For Resume Generation and Matching Tasks*, VL/HCC 2025  
<https://ieeexplore.ieee.org/document/11303517>